화학특강 (2012년 2학기)
제4장 고분자, Dendrimer 그리고 Supramolecular Chemistry

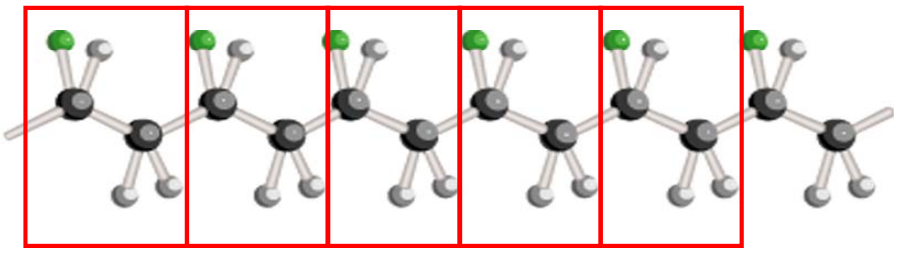
중요 학습 내용

- 고분자의 구조와 응용
- 블록공중합체와 그 응용
- Dendrimer의 구조와 응용
- Supramolecular chemistry

Polymers

Greek: "polys" ~ many
"meros"~parts

Polymers are molecules consisting of a (large) number of structural units called monomers, that are connected by covalent bonds.



The monomers are small (organic) molecules, linked together in a process called polymerization.



ethylene → polyethylene

Chang, 25.1

KNU

Molar Mass Distributions of Polymers

- Degree of polymerization (N): the number of monomers in a polymer
- Monodisperse and polydisperse
  - If all polymers in a sample have the same number of monomers, the sample is **monodisperse**.
  - If polymers in a sample consist of a various number of monomers, the sample is **polydisperse**.
- Number-average molar mass:

$$M_n = \sum_N n_N M_N$$

- Weight-average molar mass:

$$M_w = \frac{\sum_N n_N M_N^2}{\sum_N n_N M_N} = \frac{\sum_N n_N M_N^2}{M_n}$$

- Polydispersity index

$$I_p = \frac{M_w}{M_n}$$



(예제) 몰질량이 1.0×10^5 g/mol인 gelatin monomer 단백질과 몰질량이 2.0×10^5 g/mol인 gelatin dimer 단백질을 포함하는 시료가 있다. gelatin monomer의 몰분율이 0.5라고 할 때, 시료 속 gelatin 단백질의 number-average 몰질량과 weight-average 몰질량을 구하여라. 이 단백질의 polydispersity index는 얼마인가?

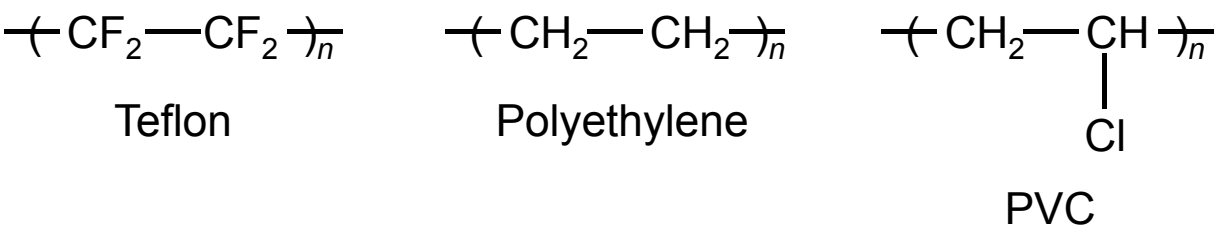
$$\text{number-average molar mass} = \sum_N n_N M_N = 1.5\times10^5 \text{ g/mol}$$

$$\text{weight-average molar mass} = \frac{\sum_N n_N M_N^2}{\sum_N n_N M_N} \cong 1.67\times10^5 \text{ g/mol}$$

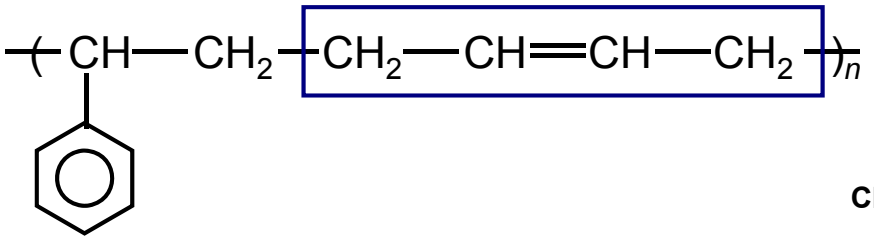
polydispersity index $I_p = \frac{M_w}{M_n} = \frac{10}{9}$

Homopolymers and Copolymers

- ***Homopolymer*** is a polymer made up of only one type of monomer



- ***Copolymer*** is a polymer made up of two or more monomers.

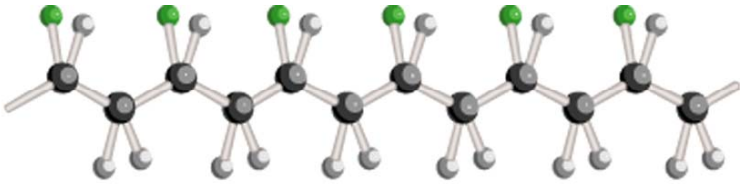


Chang, 25.1

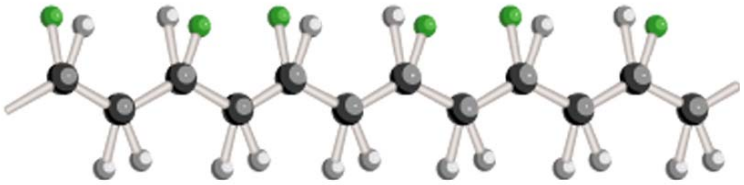
Styrene-butadiene rubber

Stereoisomers of Polymers

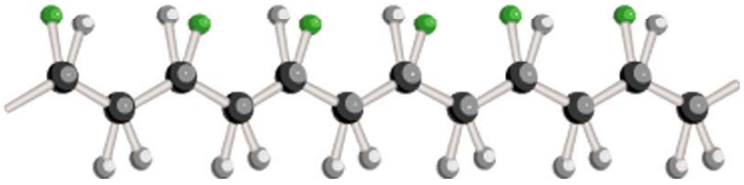
Isotactic: similar side groups always on same side of the chain. Highest melting point and great crystallinity



Syndiotactic: side groups on alternating sides of the chain



Atactic: side groups disposed at random. Rubbery,amorphous, and relatively weak



Common Polymers (1)

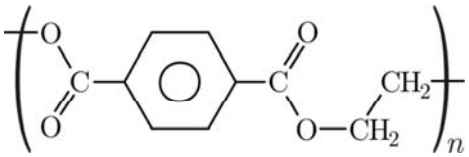
TABLE 25.1 Some Monomers and Their Common Synthetic Polymers

Monomer		Polymer	
Formula	**Name**	**Name and Formula**	**Uses**
$H_2C{=}CH_2$	Ethylene	Polyethylene $-(CH_2-CH_2)_n-$	Plastic piping, bottles, electrical insulation, toys
$H_2C{=}C(H)(CH_3)$	Propylene	Polypropylene $-(CH(CH_3)-CH_2-CH(CH_3)-CH_2)_n-$	Packaging film, carpets, crates for soft-drink bottles, lab wares, toys
$H_2C{=}C(H)(Cl)$	Vinyl chloride	Poly(vinyl chloride) (PVC) $-(CH_2-CH(Cl))_n-$	Piping, siding, gutters, floor tile, clothing, toys
$H_2C{=}C(H)(CN)$	Acrylonitrile	Polyacrylonitrile (PAN) $(CH_2-CH(CN))_n$	Carpets, knitwear

Chang, 25.2

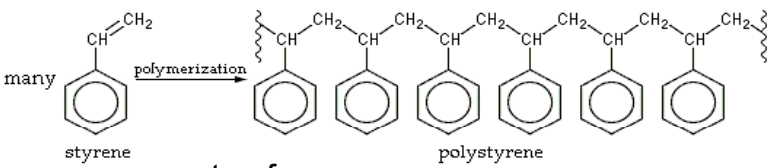
Common Polymers (2)

- Polyethylene terephthalate (PET)





- Polystyrene



styrofoam





- High-density polyethylene (HDPE) and low-density polyethylene (LDPE) and



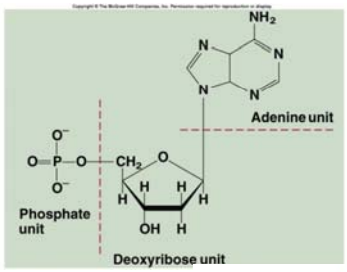




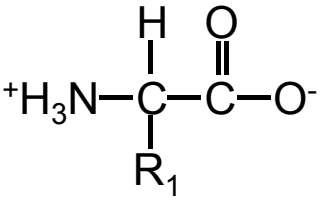


Biopolymers

- **DNA (RNA)** consists of subunits called nucleotides: sugar deoxyribose (ribose), phosphate group PO_4^- and a base.

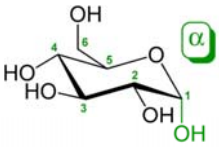


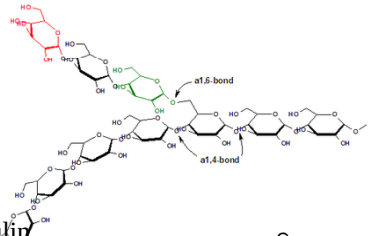
- **Proteins** are polypeptides with 20 different amino acids as the structural subunits.





- **Polysaccharides** are complex, often branched, carbohydrates, consisting of monosaccharides (simple sugars)



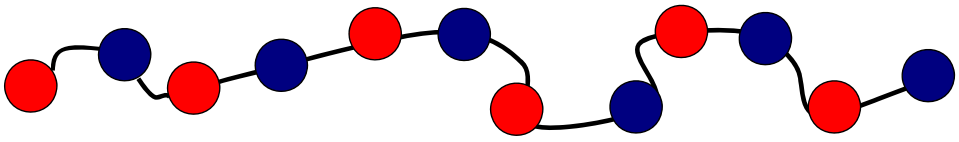


α-glucose ⇌ (insulin / glucagon) glycogen

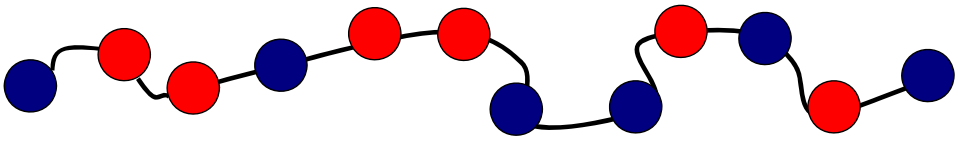
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Copolymers (공중합체)

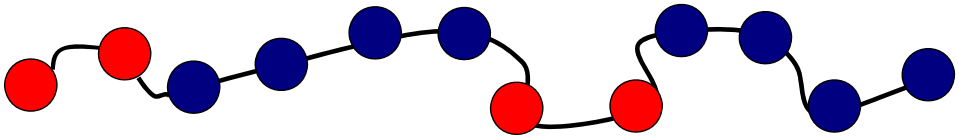
- Alternating copolymers



- Random copolymers

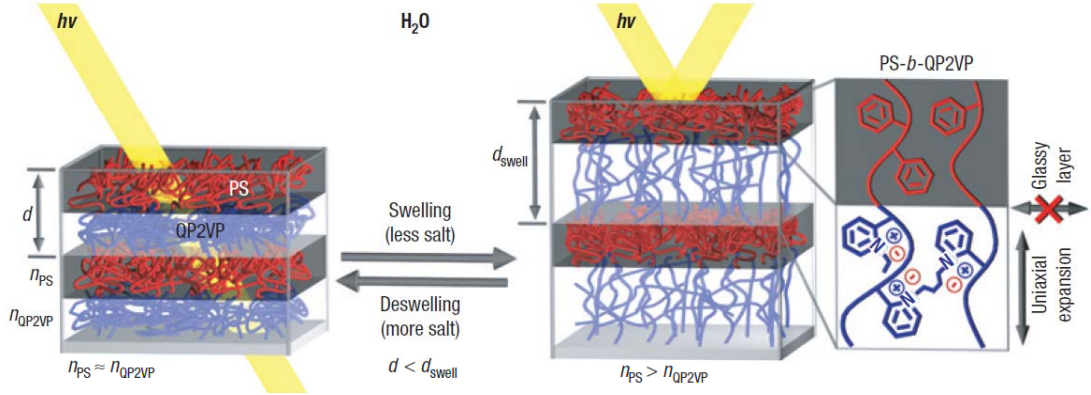


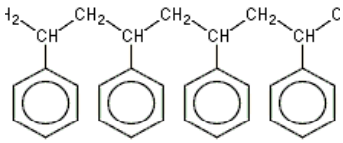
- Block copolymers



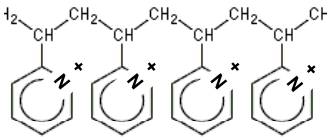
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Block-Copolymer Photonic Gel





polystyrene
(=PS)



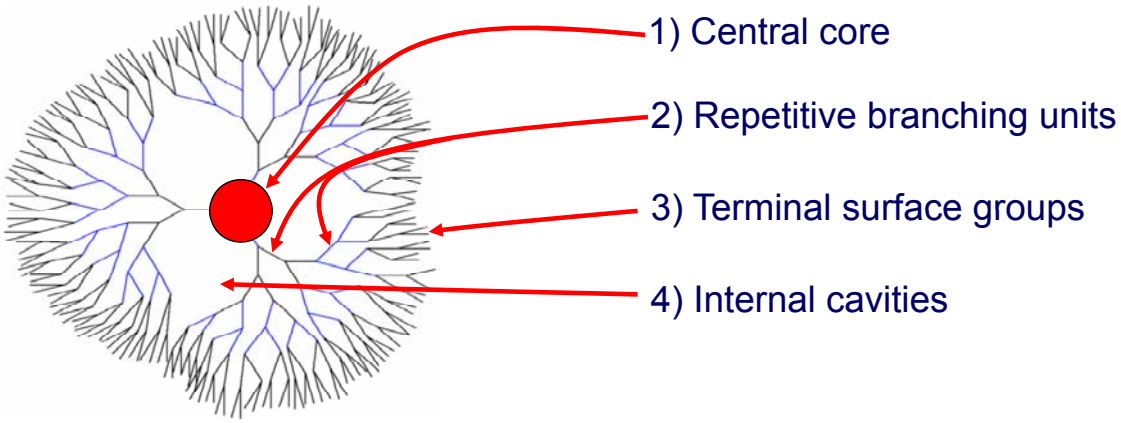
quarternized poly(2-vinyl pyridine)
(=QP2VP)



Kang et al., Nature Materials **6**, 957 (2007)

Dendrimers (1)

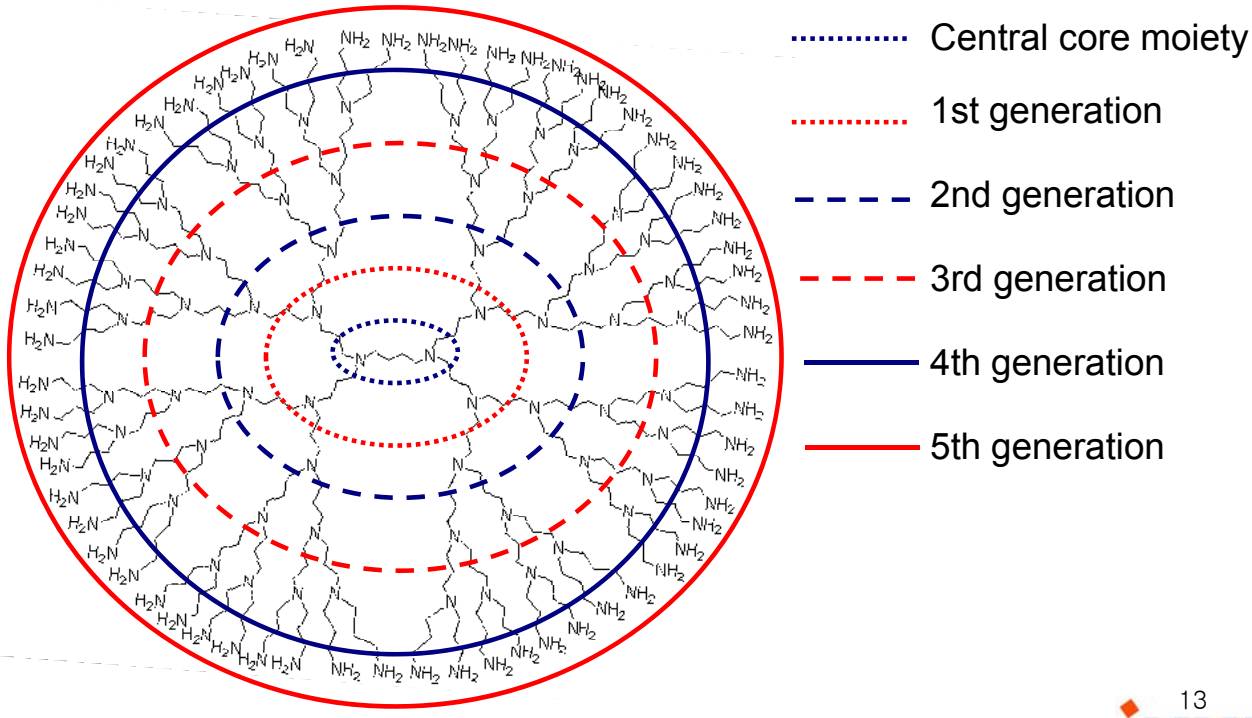
- Dendrimers: highly branched (hyperbranched) polymers that have a well defined structure. [“dendros” (Greek) ~ tree, branch]



- Divergent synthesis: from the core to the surface
- Convergent synthesis: from the surface to the core

Dendrimers (2)

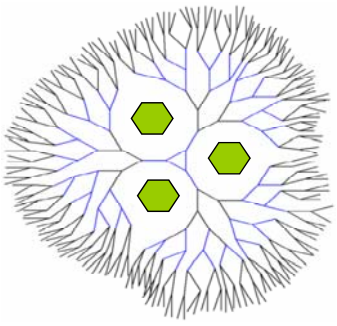
- Polyamidoamin (PAMAM) dendrimers



Dendrimers (3)

- Properties of dendrimers are dominated by functional groups on the surface of the globular molecule. For example, a dendrimer can be made water-soluble with hydrophilic surface groups
  (e.g. R-COOH $\rightarrow$ R-COO$^-$ + H$^+$)

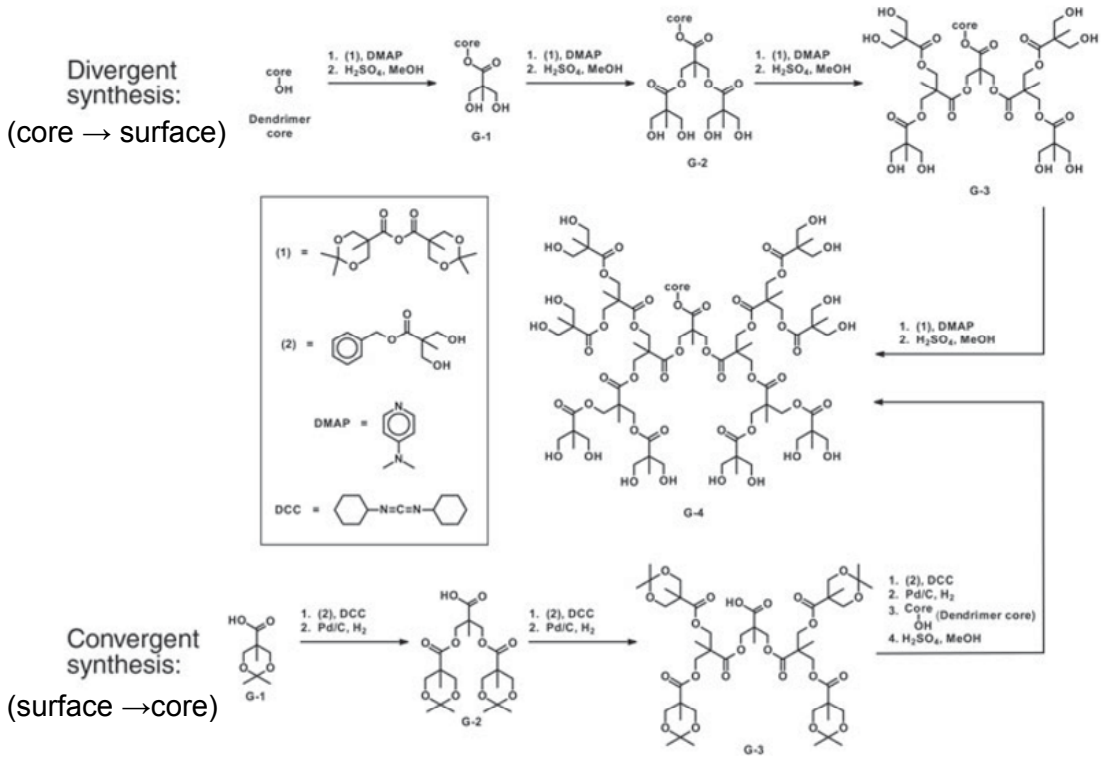
- Dendrimers with hydrophobic internal structure can act as drug carriers. Alternatively, the drug could be attached directly to the dendrimer surface groups.



= encapsulated drug

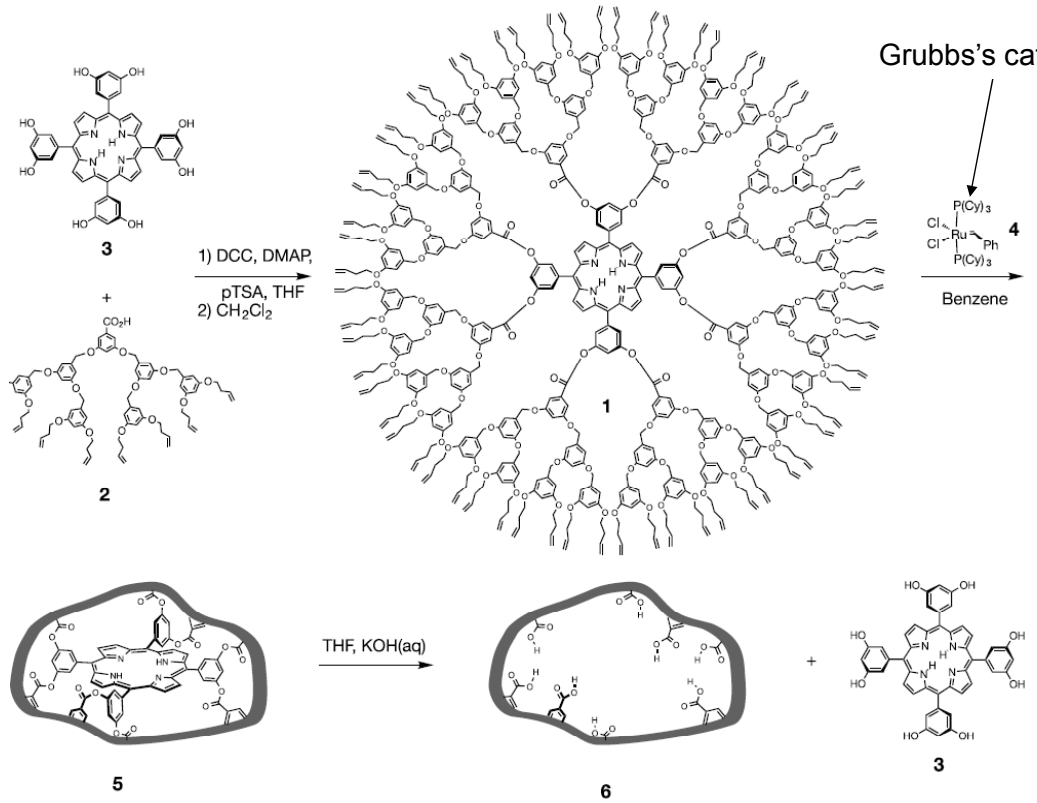
- Dendrimers have also shown promise in the development of organic LEDs: photons can be directly toward a luminescent core moiety by suitably chosen surface groups.

Dendrimer Synthesis via Divergent and Convergent Approaches



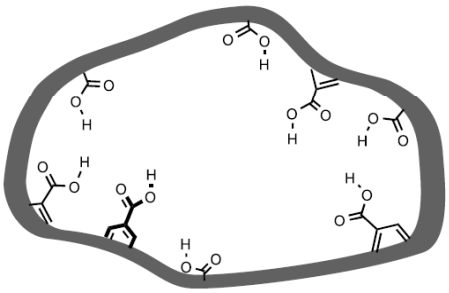
Lee et al., Nature Biotechnology **23**, 1517 (2005)

Synthetic Hosts by Molecular Imprinting inside Dendrimers (1)

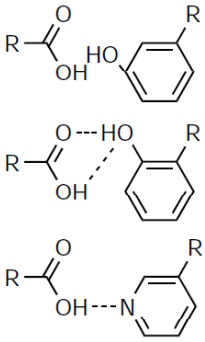


Zimmerman et al., Nature **34**, 399 (2002)

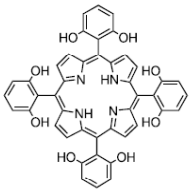
Synthetic Hosts by Molecular Imprinting inside Dendrimers (2)



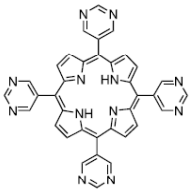
host



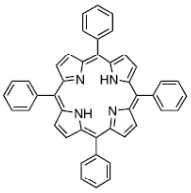
hydrogen bonding between host and guest molecules



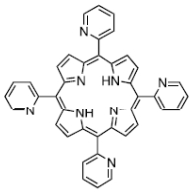
7
H_2T(2,6-OHPh)P
K_{app} = 10.0 (6.8)



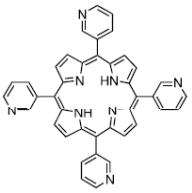
8
H_2T(3,5-pyrimidyl)P
K_{app} = 5 (0.8)



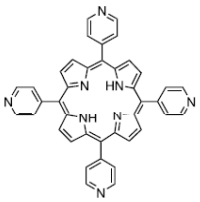
9
H_2TPP
K_{app} < 0.1 (0.0)



10
H_2T(2-pyridyl)P
K_{app} = 5 (3.0)



11
H_2T(3-pyridyl)P
K_{app} = 14 (5.8)



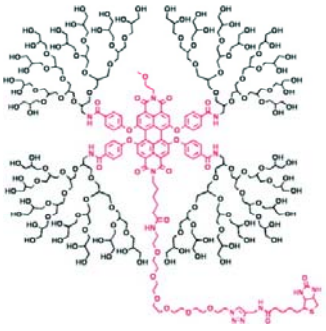
12
H_2T(4-pyridyl)P
K_{app} = 13 (5.4)

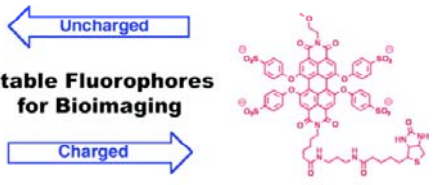
guests

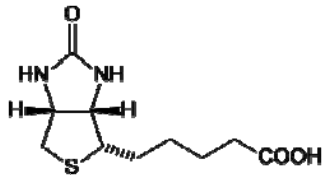
Zimmerman et al., Nature **34**, 399 (2002)

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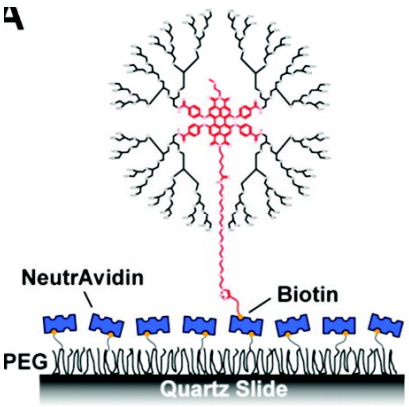
Monovalent, Clickable, Uncharged, Water-Soluble Perylenediimide-Cored Dendrimers for Target-Specific Fluorescent Biolabeling

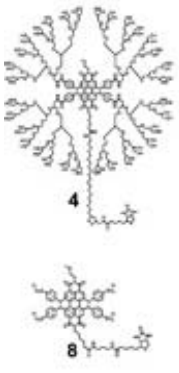


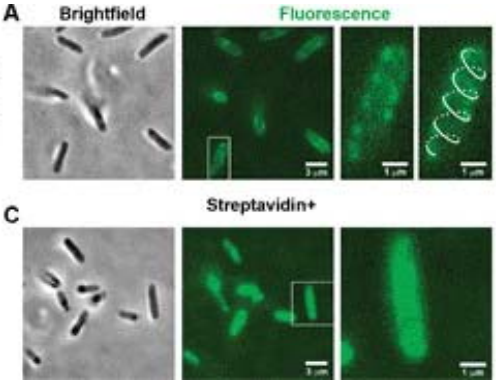




Biotin



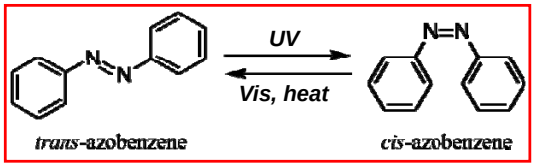




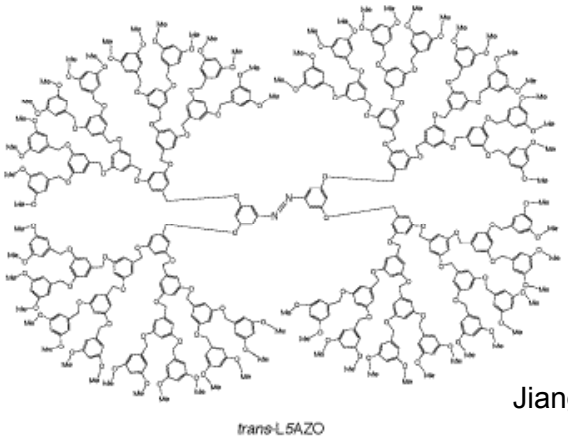
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Photoisomerization in dendrimers by harvesting of low-energy photons

- Cis-trans isomerization of azobenzene



- Photoactive dendrimer system using an IR wavenumber of 1,597 cm^{-1}: cis → trans isomerization



Jiang & Aida, Nature **388**, 454 (1997)



(예제) 앞의 실험에서 **cis → trans isomerization**이 일어나기 위해서 필요로 되는 최소한의 에너지가 **0.98 eV**라고 한다. 파수가 **1,597 cm^{-1}**인 적외선을 이용하여 **isomerization** 반응을 시키기 위해서는 적어도 몇 개 이상의 광자가 **cis-5AZO**에 흡수되어야 하는가?

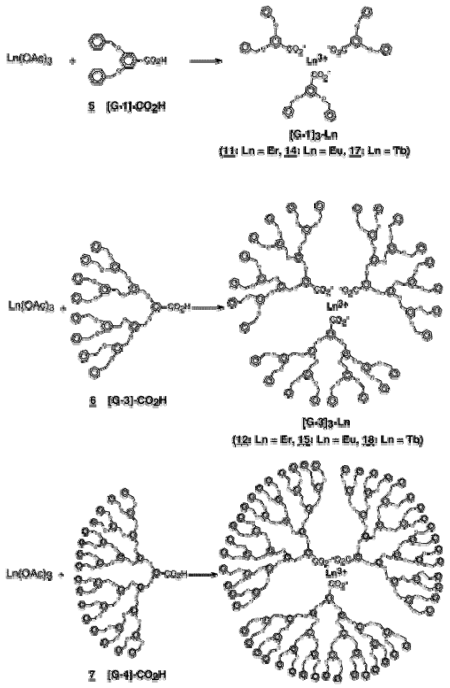
$$1\,\mathrm{eV} = 8065.51\,\mathrm{cm}^{-1}$$

$$0.98\,\mathrm{eV} = 8065.51 \times 0.98 = 7904\ \ \mathrm{cm}^{-1}$$

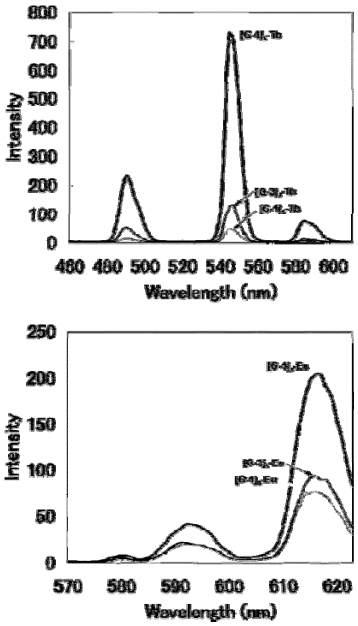
$$n = \frac{7904\,\mathrm{cm}^{-1}}{1597\,\mathrm{cm}^{-1}} = 4.94 \approx 5$$

Luminescence Properties of Lanthanide Ions through Site-Isolation and Antenna Effects

- Ln-cored dendrimer complexes



- Emission spectra of Ln-cored dendrimer complexes



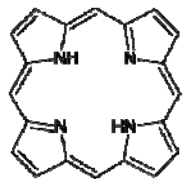
Kawa and Fréchet, Chemistry of Materials **10**, 286 (1998)

Supramolecular Chemistry

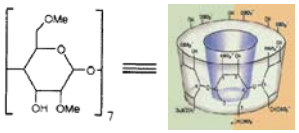
- **Supramolecular chemistry:** the area of chemistry that focuses on the noncovalent intermolecular interactions of molecules.
- Kinds of noncovalent intermolecular interactions
  - Hydrogen bonding
  - π-π stacking interaction
  - Hydrophobic interaction
  - Electrostatic interaction
    a. Charge-dipole interaction
    b. Dipole-dipole interaction
  - van der Waals interaction

Supramolecular Chemistry: Hydrophobic Interaction

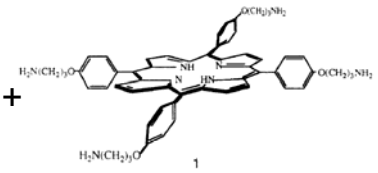
- Supramlecular complex containing porphyrin and β-cyclodextrin (β-CD) → hydrophobic interaction



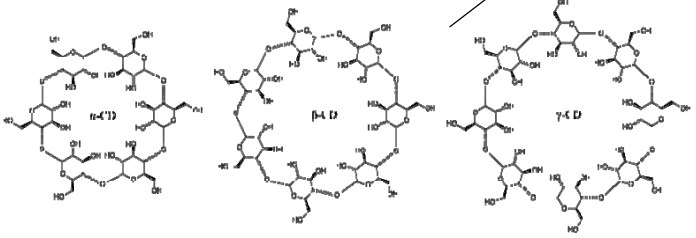
porphyrin



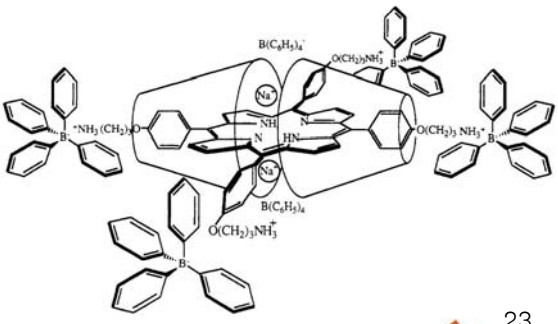
2



1



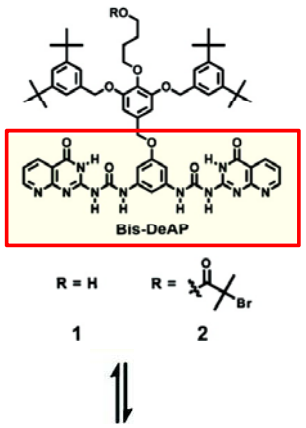
α, β, and γ -cyclodextrin

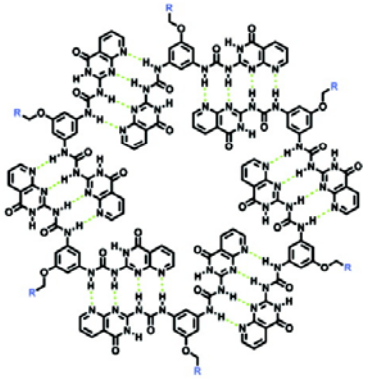


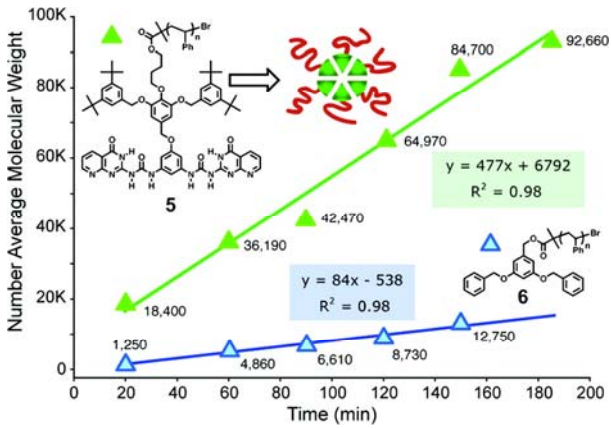
Manka and Lawrence, JACS **112**, 2440 (1990)

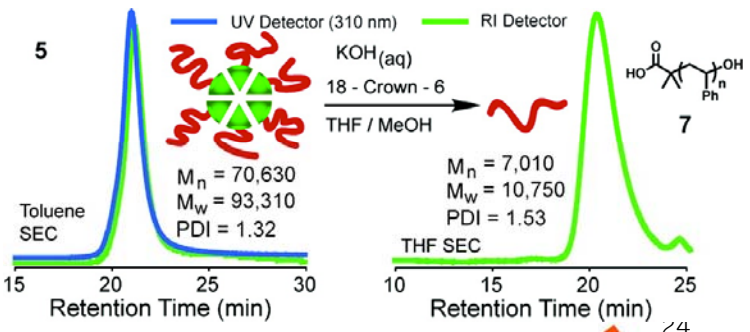
KNU

Supramolecular Star Polymers. Increased Molecular Weight with Decreased Polydispersity through Self-Assembly



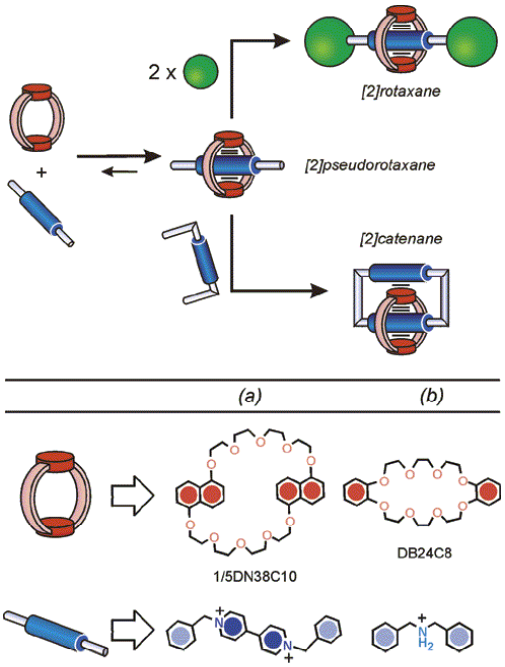






KNU

Interlocked Molecules

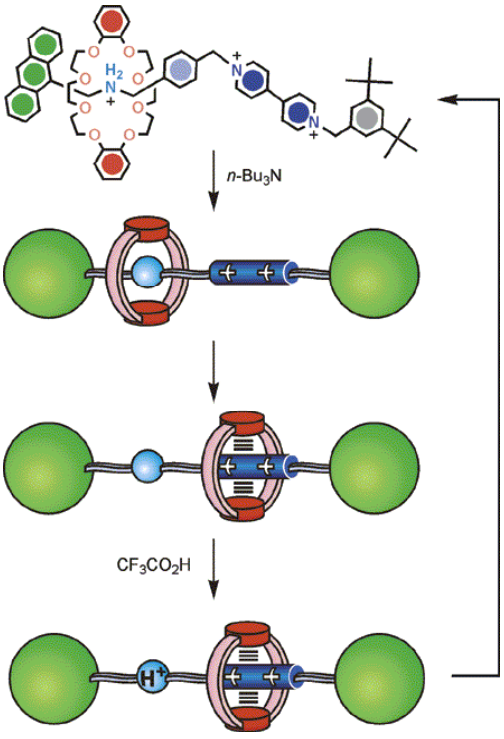


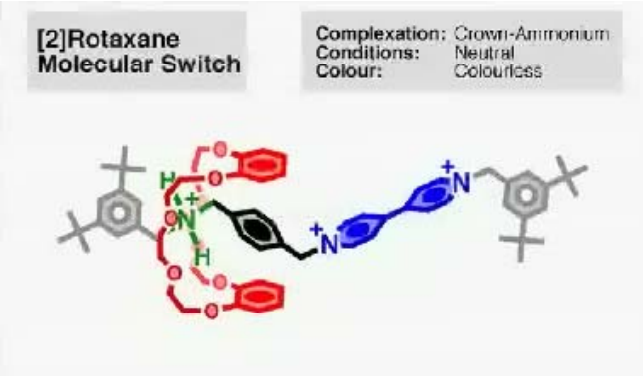
- **Rotaxane**: molecular assemblies in which a linear component is mechanically threaded through one or more cyclic molecules with the ends of the linear component terminated with bulky groups.
- **Catenane**: linked supramolecular systems consisting of cyclic subunits held by a mechanical means
- **Host guest adduct**: Interaction between π-electron rich hydroquinone and π-electron deficient pyridinium to form π stack (3.7 Å between rings). dispersive, electrostatic interactions.

Balzani et al., Acc. Chem. Res. **34**, 445 (2001)

Rotaxanes: A Molecular Shuttle

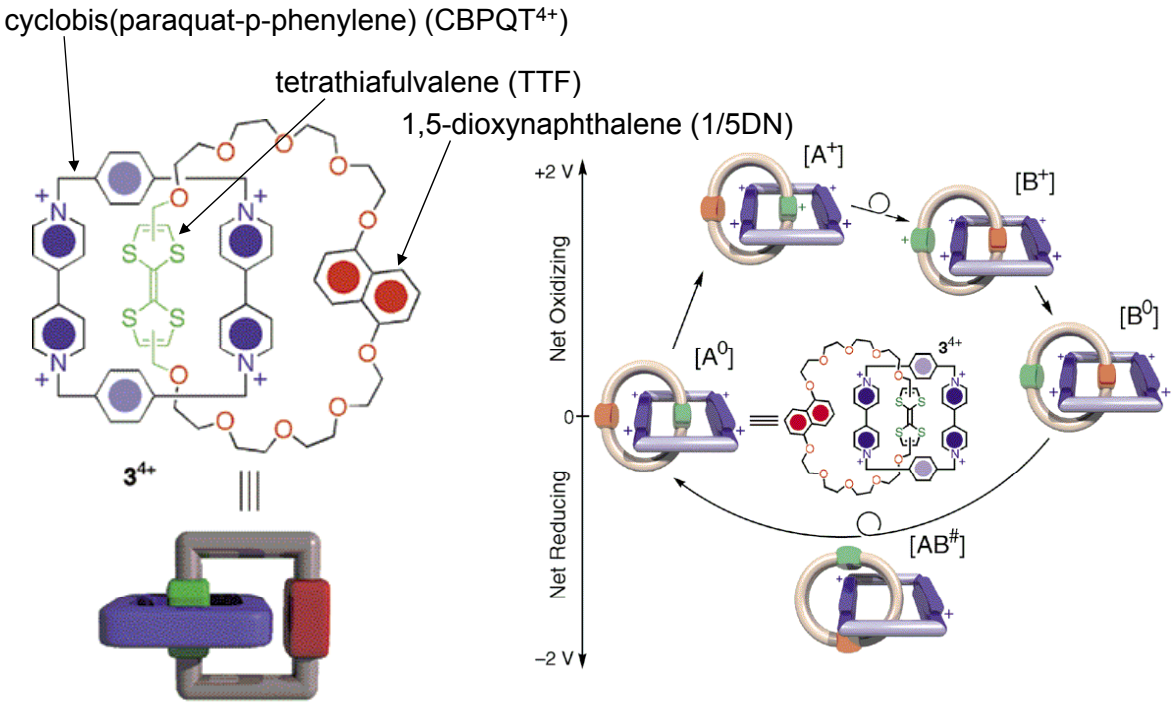
"rot" means "wheel and axa means " axis





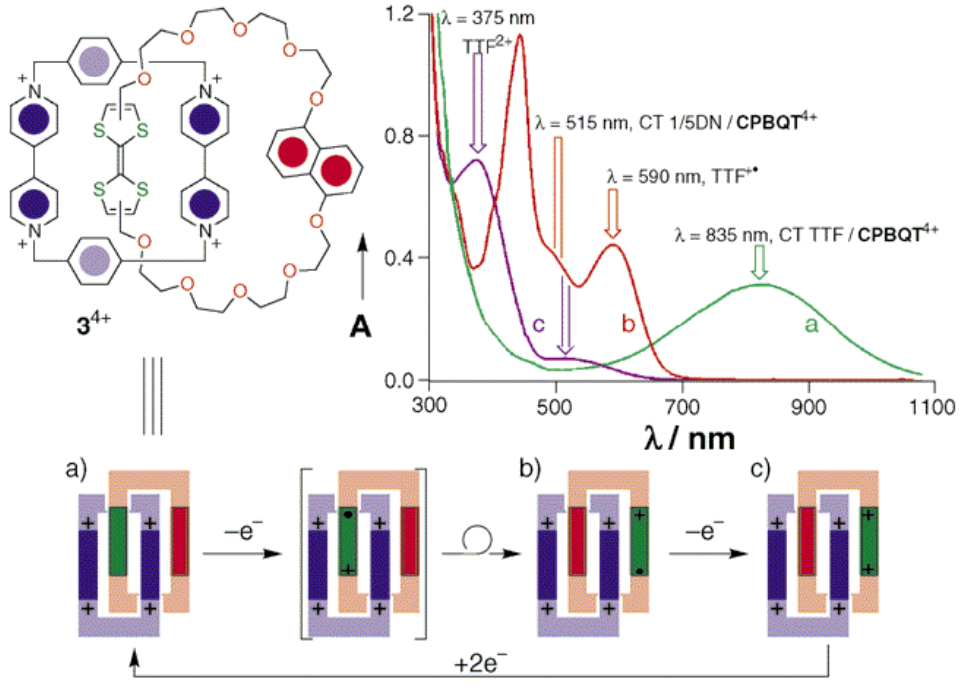
Balzani et al., Acc. Chem. Res. **34**, 445 (2001)

Catenanes: A Molecular Switch



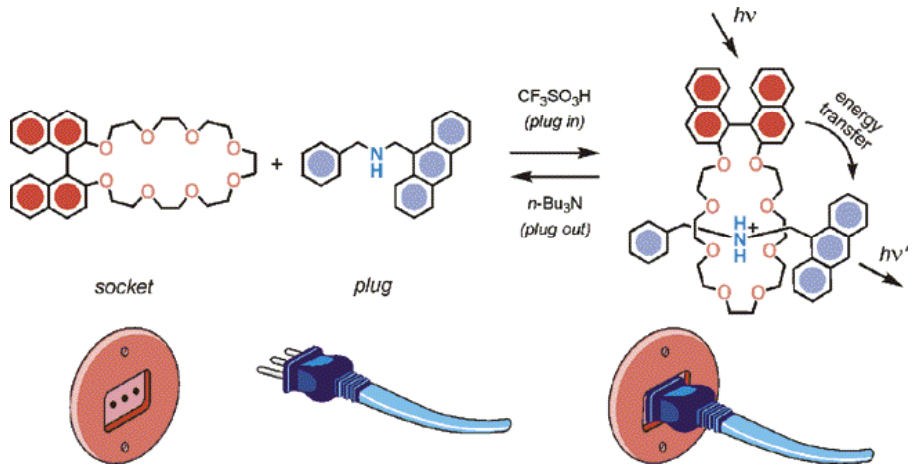
Balzani et al., Acc. Chem. Res. **34**, 445 (2001); Stoddart et al., Acc. Chem. Res. **34**, 433 (2001)

Catenanes: A Molecular Switch



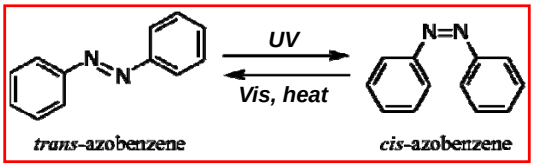
Balzani et al., Acc. Chem. Res. **34**, 445 (2001); Stoddart et al., Acc. Chem. Res. **34**, 433 (2001)

A Molecular Socket-Plug device



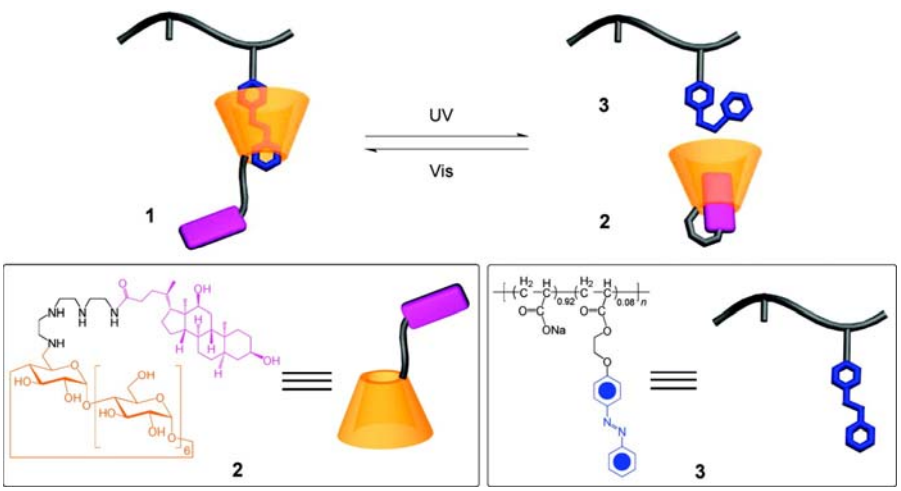
Balzani et al., Acc. Chem. Res. **34**, 445 (2001)

Azobenzene-Based Light-Responsive Hydrogel System



hydrogel

sol



Zhao & Stoddart, Langmuir **25**, 8442 (2009)